

APPLIED DATA SCIENCE(THOERY)

ASSIGNMENT NO : 2

Title : Python for Data Handling: Normalization & Standardization.

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GitHub Repo Link :

https://github.com/maheshshinde9100/SCOE_CSE_SEM_6/blob/master/AppliedDataScience/TheoryAssignments/Assignment02.py

Dataset Link :

<https://www.kaggle.com/datasets/yasserh/housing-prices-dataset>

CODE :

```
# Assignment 2 – Normalization & Standardization
# Dataset: Housing Prices (Kaggle)
# Name : Mahesh Shinde
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler, StandardScaler

print("=" * 80)
print("Libraries Imported Successfully")
print("=" * 80)

# -----
# 2. Load Dataset
# -----
df = pd.read_csv("Housing.csv")

print("Dataset Loaded Successfully")
print("=" * 80)

# -----
# 3. Clean Column Names
# -----
df.columns = df.columns.str.strip().str.lower()
```

```
print("Column Names:")
print(df.columns.tolist())
print("=" * 80)
```

```
# -----
```

```
# 4. Display Basic Information
```

```
# -----
```

```
print("First 5 Records:")
print(df.head())
print("=" * 80)
```

```
print("Dataset Shape:", df.shape)
print("=" * 80)
```

```
print("Missing Values:")
print(df.isnull().sum())
print("=" * 80)
```

```
# -----
```

```
# 5. Remove Duplicate Records
```

```
# -----
```

```
initial_rows = df.shape[0]
df.drop_duplicates(inplace=True)
final_rows = df.shape[0]
```

```
print(f"Duplicates Removed: {initial_rows - final_rows}")
print("=" * 80)
```

```
# -----
```

```
# 6. Handle Missing Values
```

```
# -----
```

```
# Fill numerical columns with mean
```

```
numerical_cols = df.select_dtypes(include=np.number).columns
df[numerical_cols] = df[numerical_cols].fillna(df[numerical_cols].mean())
```

```
# Fill categorical columns with mode
```

```
categorical_cols = df.select_dtypes(include=['object']).columns
```

```
for col in categorical_cols:
```

```
    df[col] = df[col].fillna(df[col].mode()[0])
```

```

print("Missing Values Handled Successfully")
print("=" * 80)

# -----
# 7. Select Only Numerical Columns for Scaling
# -----
numeric_data = df.select_dtypes(include=np.number)

print("Numerical Columns Selected:")
print(numeric_data.columns.tolist())
print("=" * 80)

# -----
# 8. Normalization (Min-Max Scaling)
# -----
minmax_scaler = MinMaxScaler()
normalized_data = minmax_scaler.fit_transform(numeric_data)

normalized_df = pd.DataFrame(normalized_data,
                             columns=numeric_data.columns)

print("Normalized Data (First 5 Rows):")
print(normalized_df.head())
print("=" * 80)

# -----
# 9. Standardization (Z-Score Scaling)
# -----
standard_scaler = StandardScaler()
standardized_data = standard_scaler.fit_transform(numeric_data)

standardized_df = pd.DataFrame(standardized_data,
                               columns=numeric_data.columns)

print("Standardized Data (First 5 Rows):")
print(standardized_df.head())
print("=" * 80)

print("Data Handling Completed Successfully ")

```

OUTPUT

```
Dataset Loaded Successfully
=====
Column Names: ['price', 'area', 'bedrooms', 'bathrooms', 'stories', 'mainroad', 'guestroom', 'basement', 'hotwaterheating', 'airconditioning', 'parking', 'prefarea', 'furnishingstatus']
=====
First 5 Records:
   price  area  bedrooms  bathrooms  stories  ...  hotwaterheating  airconditioning  parking  prefarea  furnishingstatus
0  13300000  7420         4          2         3  ...             no              yes         2         yes      furnished
1  12250000  8960         4          4         4  ...             no              yes         3         no       furnished
2  12250000  9960         3          2         2  ...             no              no          2         yes  semi-furnished
3  12215000  7500         4          2         2  ...             no              yes         3         yes      furnished
4  11410000  7420         4          1         2  ...             no              yes         2         no       furnished

[5 rows x 13 columns]
=====
Dataset Shape: (545, 13)
=====
```

```
=====
Missing Values:
price          0
area           0
bedrooms       0
bathrooms      0
stories        0
mainroad       0
guestroom      0
basement       0
hotwaterheating 0
airconditioning 0
parking        0
prefarea       0
furnishingstatus 0
dtype: int64
=====
Duplicates Removed
```

```
Missing Values Handled
=====
Normalized Data (First 5 Rows):
   price  area  bedrooms  bathrooms  stories  parking
0  1.000000  0.396564      0.6  0.333333  0.666667  0.666667
1  0.909091  0.502405      0.6  1.000000  1.000000  1.000000
2  0.909091  0.571134      0.4  0.333333  0.333333  0.666667
3  0.906061  0.402062      0.6  0.333333  0.333333  1.000000
4  0.836364  0.396564      0.6  0.000000  0.333333  0.666667
=====
Standardized Data (First 5 Rows):
   price  area  bedrooms  bathrooms  stories  parking
0  4.566365  1.046726  1.403419  1.421812  1.378217  1.517692
1  4.004484  1.757010  1.403419  5.405809  2.532024  2.679409
2  4.004484  2.218232  0.047278  1.421812  0.224410  1.517692
3  3.985755  1.083624  1.403419  1.421812  0.224410  2.679409
4  3.554979  1.046726  1.403419 -0.570187  0.224410  1.517692
Data Handling Completed Successfully
```